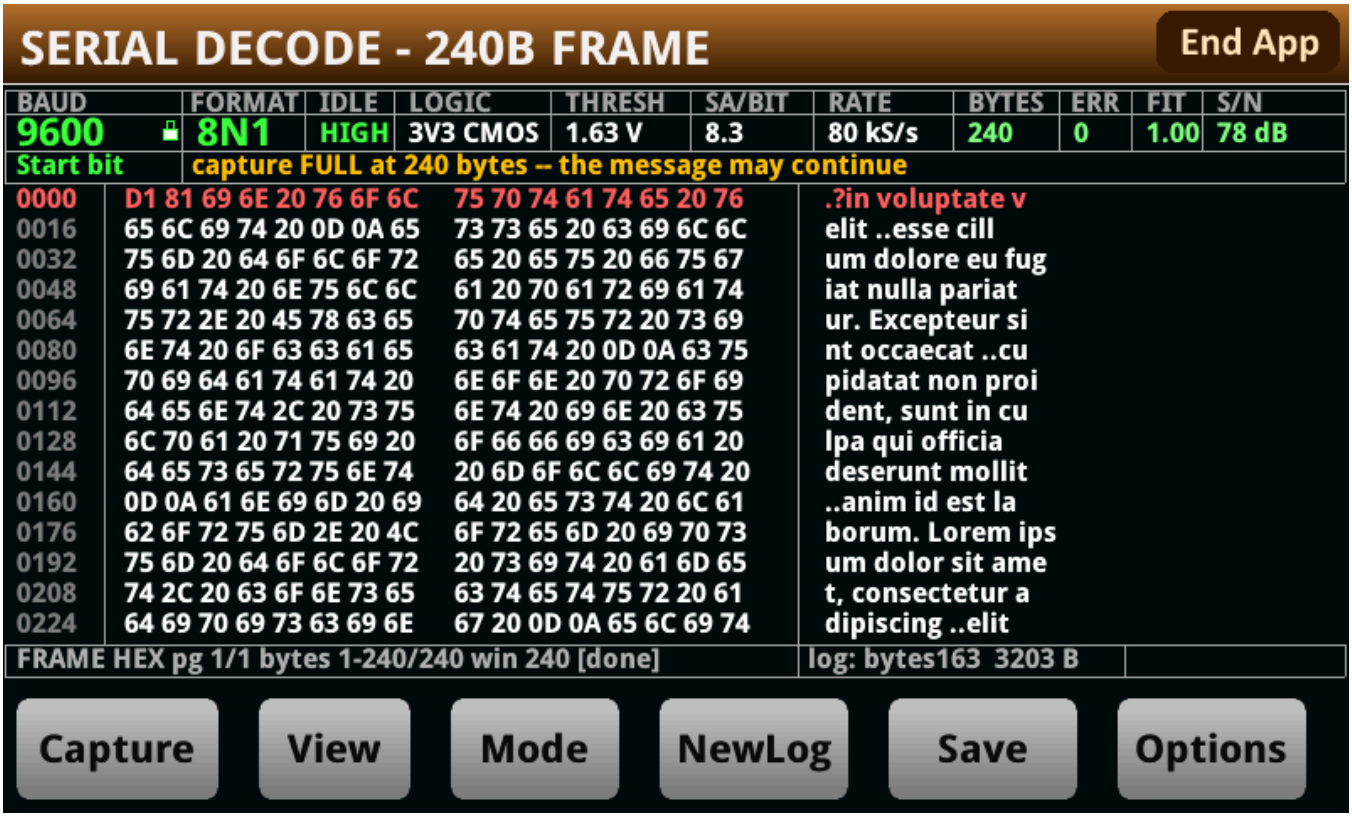
Serial Decode — Keithley DMM6500, DAQ6510, DMM7510, SMU2461

A UART decoder that runs on a Keithley bench instrument. It digitizes the line with the instrument's own digitizer, automatically recovers the baud rate, frame format and idle polarity from the signal, and shows the bytes on the front panel as text or hex. No host computer, terminal, or logic analyzer needed, no setting baud rates, stop bits, data bits, parity, etc - just one probe, and one button press. Press **Capture** before the device under test has started talking and the capture **waits for it** — up to two minutes — so a board that only speaks when you power it on is caught from its first byte rather than recorded as silence. The signal must swing by at least 0.33V, lie between -10V and +10V, have no more than 30% noise, and under 15% clock jitter. Any baud rate will work from 110 to 250,000 inclusive - even non standard ones. Any DC offset is fine so long as these conditions are met.

Ian Ameline · **version 1.45** · MIT licence (see LICENSE)



The instrument's own front panel, no host attached: 239 bytes of a 240-byte capture at 9600 baud 8N1, no framing errors, S/N 74 dB. The padlock on the BAUD cell means the rate is pinned here rather than re-detected each capture.

docs/MANUAL.md	using it: hooking up, the screen, the buttons, what it copes with, where it fails
docs/REFERENCE.md	every measured number, and the firmware limits behind them
docs/BENCH.md	the harness: the seeded sweep, the release gate stage by stage, reproducing a failure

Ships as `Serial_Decode.tspa`, installed from a USB key through the instrument's own **Manage Apps** screen. Written in TSP — Lua **5.0.2** embedded in the firmware.

Installing it

The app ships as a single file, `Serial_Decode.tspa`. It installs from a USB key through the instrument's own **Manage Apps** screen — no host software, no driver, nothing to compile. The archive is plain ASCII: a `loadscript Serial_Decode` wrapper around the six modules and an icon, so it can be read and diffed like any other source file.

1. Copy the archive to a USB key. Put `Serial_Decode.tspa` in the **root** of a FAT-formatted key. Leave the name as it is.

> **Check the copy before you install it.** A truncated or corrupted archive does not fail politely — it **bluescreens the instrument**, repeatably, and the crash says nothing about why. Observed on a key whose filesystem had become inconsistent; recopying the identical file fixed it outright.

2. Put the key in the front-panel USB port.

3. Install it from the front panel, through **MENU ▸ Manage Apps**, which copies the app into the instrument's internal memory. The exact wording of that screen varies between models and firmware revisions; it is the one that lists the apps on the key.

4. Run it from the instrument's **Apps** menu. It comes up idle, showing `ready -- press Capture`, and measures nothing until you ask it to — so launching never waits on an acquisition. Press **Capture** and a line already carrying traffic decodes immediately; a pin with nothing on it reports `line is idle (no transitions)`.

5. Leave the key in, or take it out. With a key in the slot the app appends decoded bytes to a log and the `Save` button writes a report; with no key it captures and decodes exactly the same in every mode, says `log: no USB key`, and reports a recording as `NOT recorded` rather than naming a file. The key may be removed and replaced while the app runs.

Hooking up the probe

`INPUT HI` and `INPUT LO` on the front terminals. `LO` to the ground of the circuit you are probing, `HI` to the line you want to read — everything is measured at HI relative to LO, so without a shared ground the readings mean nothing.

- **Keep the line between −10 V and +10 V.** The app fixes the meter on its 10 V range. 3.3 V CMOS, 5 V TTL and a 6 V LIN line tapped to ground need nothing in front of them. For 12 V, use a 2:1 divider — and on an open-drain bus pick resistors big enough not to load it down.
- **A DC offset does not matter.** The app measures the two levels on your line and puts its threshold between them, and it works out the idle polarity from the signal itself.
- **Do not put a differential bus across HI and LO.** RS-485, CAN and LIN want a transceiver first, or tap one side of the pair against ground.
- **A signal whose band straddles ground is decoded inverted, deliberately** — see docs/BETA.md, which lists the behaviours that look like defects and are not.

Then: press **Capture** while the device is talking, and **View** to switch between hex and text. The manual covers the rest of the panel, the bit rates it reaches, and what a marginal signal looks like.

Building the archive yourself

Only needed if you have changed anything in `tsp/`:

```
python3 tools/package_tspa.py        # concatenates tsp/*.tsp into Serial_Decode.tspa
```

Everything else

docs/PROJECT.md	what to know before trusting it, the measured endurance figures, which instruments it runs on, how to report a bug, and the release gates
tools/README.md	the harness: offline suites, the bench sweep, the release gate